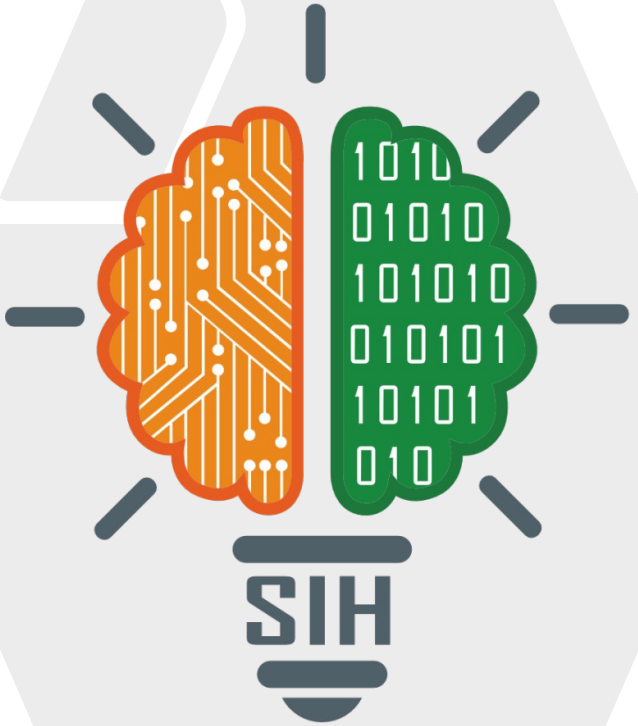
SMART INDIA HACKATHON 2026

SCOURGE

Ask the bank, not the paper.



- **Problem Statement ID –** SIH26265
- **Problem Statement Title –** Streamline of Security Deposit / FDR Portal
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** 199322
- **Team Name (Registered on portal) –** Arnavax#

SCOURGE

SCOURGE — ASK THE BANK, NOT THE PAPER

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Proposed Solution (Describe your Idea/Solution/Prototype)

An institution proves what its bank's own server says about one FDR. AICTE sees only that record.

Detailed explanation of the proposed solution

AICTE sends a **single-use challenge** for one FDR. The institution logs into its own net banking and proves the deposit page inside one TLS session whose keys are **split with AICTE's verifier** (MPC-TLS).

How it addresses the problem

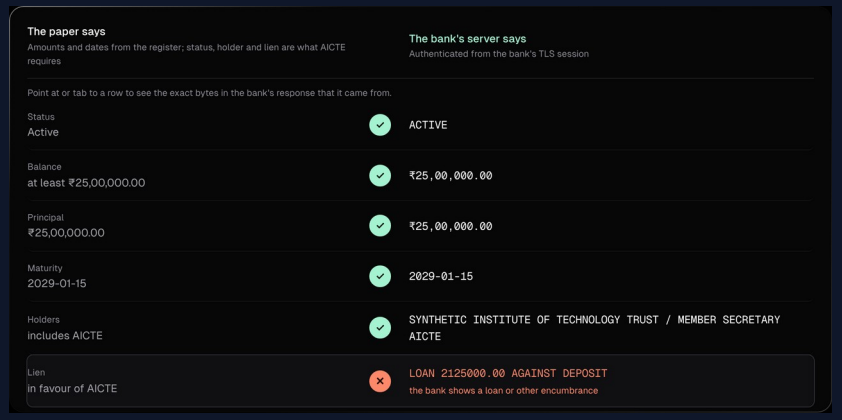
Paper is checked once and trusted for 10 years. Scourge **re-checks the bank any time and before every release**: a loan, closure or fake FDR shows up as a **Discrepancy**.

Innovation and uniqueness of the solution

No bank API, no agreement, no AA licence. The institution cannot forge it (a fake bank is rejected) and AICTE never sees the PAN or other accounts. None of the ~27 public SIH26265 builds we found does this.

LIVE PROTOTYPE · SYNTHETIC BANK

The bank's server contradicts the paper.



× HIDDEN LOAN CAUGHT

Every value traces to the bank's own bytes

The complete FDR lifecycle in one portal (all five stages run in the prototype)

1 FILE — Institution files its FDR claim online

→ **2 PROVE** — Bank-session proof on AICTE's challenge

→ **3 MONITOR** — Maturity alerts; re-check every 30 days

→ **4 DECIDE** — Officer accepts or escalates; audit log

→ **5 RELEASE** — Only with a fresh bank confirmation

Why now: **₹201.42 crore of FDRs "do not appear in IDFC's records"** (CBI chargesheet, Oct 2026); AICTE joint FDRs found encashed or loaned against without consent.

SCOURGE

TECHNICAL APPROACH

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Technologies to be used

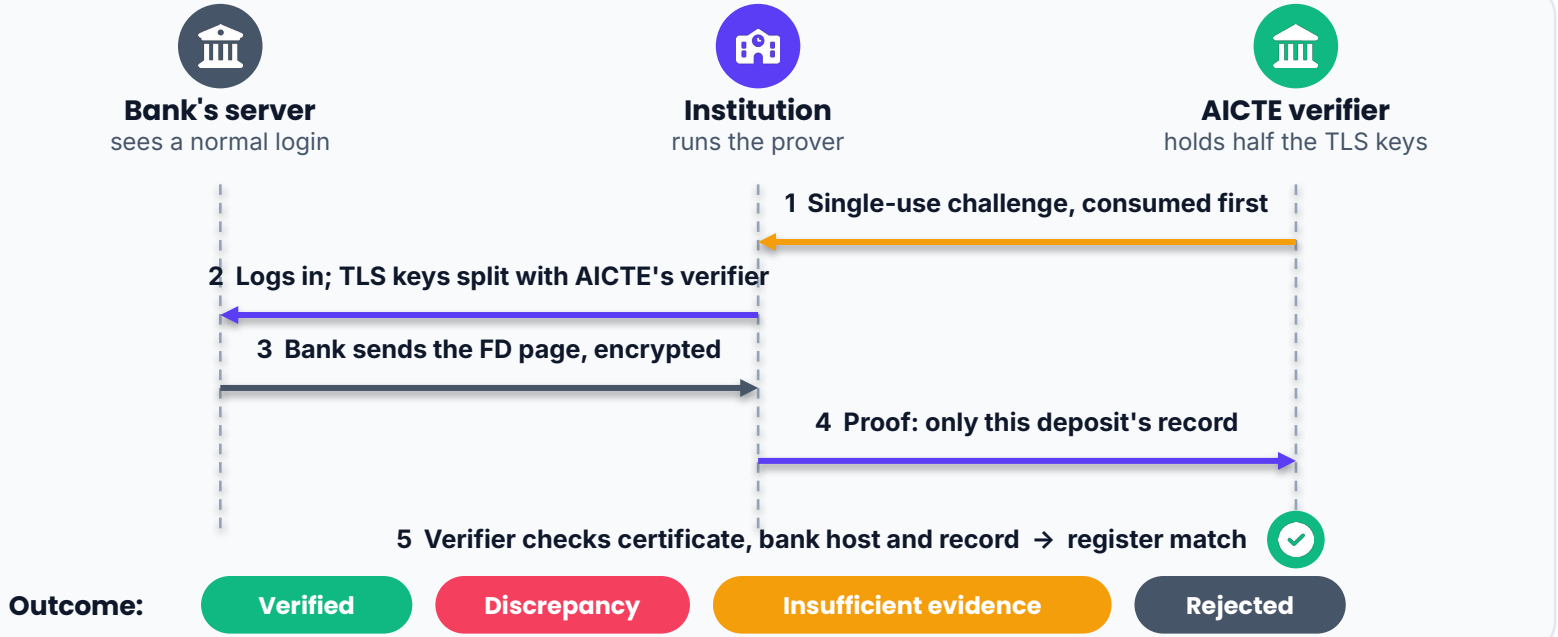
Proof layer
TLSNotary MPC-TLS · Rust
Selective disclosure · rustls

Backend
Python · FastAPI · SQLite
Separate verifier process

Frontend
Server-rendered HTML · JS
Live 3-party proof view

Ops & QA
nginx TLS · systemd
pytest · Playwright · audits

Methodology and process for implementation (Flow Charts/Images/ working prototype)



WATCH THE 90-SECOND DEMO



Click to play here, or scan / click the QR for YouTube
Real proofs on a synthetic bank

SCOURGE

FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

0.58 s
per bank proof
median of 60 runs · synthetic bank

13 / 16
real TLS sessions, 6 Indian banks
public pages, not FD pages

102 / 102
automated tests pass
local run · 2 deploy-only skipped

₹0
software licences
one small server [estimate]

- **Technical:** Open-source TLSNotary (MIT/Apache-2.0); runs on one small server
- **Operational:** Institution uses its own login; no bank integration needed
- **Viability:** Adopted by an AICTE circular; covers the existing FDR book

Potential challenges and risks

Strategies for overcoming these challenges

Potential challenges and risks	Strategies for overcoming these challenges
Bank page hides liens, or bytes are withheld	Insufficient evidence, never Verified; release stays blocked
Fake bank site or a replayed old proof	Rejected: bank certificate + single-use, 15-min challenge
Fraud inside the bank's own records	Out of scope by design: Scourge trusts the bank's server, not paper
Bank needs browser-bound session / TLS 1.3	Browser-extension prover; bank marked unsupported until then

NEXT GATE Prove a logged-in FD page at a real bank with our own account. Pass on one bank = launch there. Bank terms and AICTE acceptance go to legal review via the SPOC.

SCOURGE

IMPACT AND BENEFITS

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Potential impact on the target audience

AICTE RB officers
A dated proof from the bank in seconds, not a letter in weeks

AICTE finance
The whole FDR book is re-checkable, before every release

Institutions
One short proof from their own net banking; nothing else revealed

Students
The deposit that protects them is checked across all 10 years

Benefits of the solution (social, economic, environmental, etc.)

TODAY	WITH SCOURGE
Branch letter or uploaded screenshot	**The bank's own server, proven**
Checked once, trusted for 10 years	**Re-checked any time, before every release**
Weeks per confirmation	**Under a second per proof (synthetic bank)**
Hidden loans found at release, or never	**A Discrepancy, with the bank's bytes**

TESTED ON 180 FRESH DEPOSITS
Random amounts, dates, formats · real proofs · synthetic bank

119 / 119	bad deposits the upload portal approved
0 / 119	bad deposits Scourge verified
0 / 61	honest deposits Scourge accused

Social
students' deposits stay intact

Economic
stops loss before release

Environmental
no paper letters or couriers

Governance
dated, exportable evidence

SCOURGE

RESEARCH AND REFERENCES

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Details / Links of the reference and research work

The problem is real

CBI: ₹201.42 cr of FDRs not in IDFC's records (ThePrint, 7 Oct 2026)
theprint.in › cbi-says-haryana-pollution-board...

CBI on AICTE: joint FDRs encashed or loaned without consent
deccanherald.com › greater-transparency-aicte...

Regulation

AICTE guidelines 2024-27: cl. 6.2(g) 10-year release; cl. 11.8
indiankanoon.org/doc/173919027

Account Aggregator users must be regulated FIUs (DFS)
financialservices.gov.in › account-aggregator...

Technology

TLSNotary: MPC-TLS proofs of web data, MIT / Apache-2.0
tlsnotary.org

NeSL e-BG for security deposits: new instruments only (WDRA)
wdra.gov.in › ebg_requirement-27-08-2025

PROJECT RESOURCES

Demo video (90 s): youtu.be/8HoplhFOGCg
Prototype: Scourge v1.0.0-rc1 · 102 / 102 tests · synthetic bank
Code + evidence: github.com/Lakki007/Arnavax-Scourge-SIH26265

Scourge vs existing approaches

	Upload & approve	Bank letter	Account Aggregator	NeSL e-BG	Scourge
Uses the bank's own data	✗	✓	✓	✓	✓
No bank API or agreement	✓	✓	✗	✗	✓
Institution cannot forge it	✗	~	✓	✓	✓
Covers existing FDR book	✓	✓	✓	✗	✓
Re-check in seconds	✗	✗	✓	✓	✓
Shown on a real bank today	✓	✓	✓	✓	Next gate

~ a bank official can still forge a letter. Real-bank proof is our next gate (slide 4).